

# Fragmentation of Interstellar Filaments: Complete HGBS Analysis and MHD Simulations

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## ABSTRACT

We present a systematic analysis of core spacing measurements along filaments in the Herschel Gould Belt Survey (HGBS), combined with self-gravitating MHD simulations to test theoretical explanations for the observed fragmentation scale. Our sample includes 8 high-confidence HGBS regions (5,069 cores), with distances updated to Gaia DR3-based measurements from [Zhang et al. \(2023\)](#). The weighted mean core spacing is  $0.279 \pm 0.009$  pc, corresponding to  $2.79\times$  the characteristic filament width of 0.10 pc. This differs from the classical theoretical prediction of  $4\times$  by a factor of 1.4.

We examine two primary explanations. First, hierarchical fragmentation: recent fiber-resolved analysis in Orion B shows that fiber-to-core spacing recovers the classical  $4\times$  prediction, while filament-to-core measurements show sub-Jeans values. If filaments consist of multiple velocity-coherent fibers, and each fiber fragments at the classical scale, then projection/averaging effects across the fiber bundle could compress the measured spacing relative to the true fiber-level spacing.

Second, magnetic tension along filaments: for longitudinal magnetic fields with plasma  $\beta \sim 1\text{--}3$ , the full numerical solution predicts  $\lambda/W = 2.4\text{--}3.2$ , overlapping with our measurement. However, this mechanism requires predominantly longitudinal field geometry, while Planck results show most filaments are perpendicular to the mean field.

We present self-gravitating MHD simulations using Athena++ spanning the (Mach, plasma  $\beta$ ) parameter space relevant to molecular cloud filaments. The simulations reveal three distinct fragmentation regimes and reproduce the expected fragmentation timescales from linear MHD stability theory. We additionally present: (1) a comprehensive supercritical filament campaign (654 simulations) spanning  $f = 1.1\text{--}3.0$ ,  $\beta = 0.3\text{--}5.0$ , and  $\mathcal{M} = 0.5\text{--}3.0$ ; (2) a Definitive Transition Campaign (540 simulations) mapping the stable-unstable boundary for longitudinal B-fields; (3) three validation campaigns (83 simulations) testing resolution convergence (verified with matched problem generator:  $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$ ), initial condition sensitivity, and equation of state effects; (4) a Field Geometry Campaign (314 simulations) testing perpendicular and oblique field geometries, near-critical fragmentation, and adiabatic EOS; and (5) targeted re-runs with extended 6-hour wall-clock timeouts (25 simulations, April 2026) demonstrating that all DTC STABLE classifications at  $\beta = 0.3$ ,  $\mathcal{M} = 1$  were timeout artifacts. The combined 1616 simulations provide the most extensive numerical test of filament fragmentation to date, revealing a robust power-law scaling  $t_{\text{frag}} \propto f^{-0.39}$  ( $r^2 = 0.999$ ) and demonstrating that fragmentation times are independent of Mach number and nearly independent of magnetic field strength for longitudinal field geometry.

**Key words:** stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

## EXECUTIVE SUMMARY OF LIMITATIONS

**What the simulations CAN test:** Our MHD simulations measure fragmentation timescales ( $t_{\text{frag}}$ ) and stability boundaries in the ( $f, \beta, \mathcal{M}$ ) parameter space. The Field Geometry Campaign (314 simulations) demonstrates robust longitudinal beading in near-critical filaments ( $f \approx 1.0\text{--}1.2$ ) with 100% fragmentation at  $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$ , while perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields.

**What the simulations CANNOT directly test:** Our

supercritical filament simulations ( $f \geq 1.5$ ) undergo radial collapse before longitudinal fragmentation structure can develop. Analysis of all 654 supercritical simulations yielded zero detections of longitudinal beading—all show pure radial collapse with longitudinal density variation  $< 0.02\%$ . This prevents direct numerical measurement of the fragmentation spacing  $\lambda/W$  in the supercritical regime. The calibration  $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}$  comes from earlier near-critical simulations (where beading does occur), not from the supercritical campaign presented here.

**Key uncertainties:** (1) *DTC timeout artifacts:* The Definitive Transition Campaign used 600-second wall-clock timeouts and identified STABLE configurations. The supercritical campaign with longer timeouts (7200–10800 s) showed that many of these eventually fragment. Targeted re-runs with 6-hour timeouts (completed April 2026) confirmed that all DTC STABLE classifications at  $\beta = 0.3$ ,  $\mathcal{M} = 1$  were timeout artifacts. (2) *Resolution convergence:* Explicitly verified with matched problem generator, showing  $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$  (mean difference  $7.2\% \pm 1.6\%$ ), confirming that  $128^3$  is adequate for fragmentation classification. (3) *Statistical limitations:* The pairwise median spacing statistic has poorly characterized sampling properties for small-N filaments, and distance uncertainties (3–5%) propagate directly to  $\lambda/W$  measurements.

**Regime-dependent behavior:** The simulations reveal a critical regime change at  $f \approx 1.2$ – $1.5$ : near-critical filaments ( $f \lesssim 1.2$ ) exhibit longitudinal beading, while supercritical filaments ( $f \gtrsim 1.5$ ) undergo radial collapse. This is not a numerical artifact but a real physical result with important implications for interpreting HGBS observations. We present limitations upfront rather than burying them in discussion to enable honest assessment of what can and cannot be concluded from this work.

## 1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al., 2010). Two key observational results emerged: (1) a characteristic filament width of  $\sim 0.1$  pc across diverse environments (Arzoumanian et al., 2011), and (2) a critical mass-per-unit-length of  $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$  (Ostriker, 1964).

Classical fragmentation theory (Inutsuka & Miyama, 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately  $4\times$  the filament width. For the characteristic HGBS filament width of  $0.10$  pc, this predicts a core spacing of  $\sim 0.4$  pc. Original HGBS analyses measured core spacings of  $\sim 0.2$  pc—nearly a factor of two smaller than the theoretical prediction. However, with the adoption of Gaia DR3-based distances (Zhang et al., 2023), the measured spacings have increased to  $\sim 0.28$  pc, reducing the discrepancy with theory to a factor of  $1.4$ .

Two primary explanations have been proposed for this discrepancy. First, **hierarchical fragmentation:** high-resolution studies revealed that filaments consist of velocity-coherent fibers (Hacar et al., 2013), and recent work in Orion B (Yang et al., 2024) showed that while filament-to-core spacing is  $\sim 2$ – $3\times$ , fiber-to-core spacing recovers the classical  $4\times$  prediction. This suggests the apparent sub-Jeans spacing at the filament level may reflect projection/averaging effects from measuring across hierarchical structures rather than a true modified fragmentation wavelength.

Second, **magnetic tension:** longitudinal magnetic fields resist bending during fragmentation, preferentially stabilising long-wavelength perturbations and shifting the most unstable mode to shorter wavelengths (Nakamura et al., 1993). For equipartition-strength fields ( $\beta \sim 1$ – $3$ ), this mechanism predicts  $\lambda/W \approx 2.4$ – $3.2$ , overlapping with HGBS observations.

In this paper, we present the first complete HGBS analy-

sis across all 8 high-confidence regions, combined with self-gravitating MHD simulations to test these competing explanations.

## 2 OBSERVATIONAL ANALYSIS

### 2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated from the original HGBS catalogue estimates to the latest Gaia DR3-based measurements from Zhang et al. (2023). The original HGBS catalogues (Könyves et al., 2015, 2020; André et al., 2016; Ladjelate et al., 2020) adopted distances ranging from 130–260 pc based on the best available estimates at the time (2010–2015). Since then, Gaia DR3 parallaxes have enabled substantial distance revisions for several regions, particularly those at larger distances where parallax measurements are more precise.

**Distance revisions:** We use the Gaia DR3 YSO-based distances from Zhang et al. (2023) (Table 4), which have typical uncertainties of 3%–5%. The most significantly affected regions are: Serpens (260  $\rightarrow$  458 pc, +76%), Aquila (260  $\rightarrow$  436 pc, +68%), and Orion B (260  $\rightarrow$  386 pc, +48%). Taurus shows a small decrease (140  $\rightarrow$  135 pc, -4%), while Ophiuchus is nearly unchanged (130  $\rightarrow$  137 pc, +5%). Core spacings scale linearly with distance, so these revisions proportionally increase the measured spacings for affected regions.

**Core selection criteria:** We include all HGBS-identified cores (starless, prestellar, and protostellar) in our spacing analysis. The HGBS catalogues use a consistent core extraction and classification framework across all regions (Könyves et al., 2015, 2020). While including unbound starless cores and protostellar sources may introduce some bias, we adopt the inclusive HGBS definition for two reasons: (1) fragmentation processes produce a continuum of core boundness states, and (2) restricting the sample to prestellar cores only would reduce sample sizes significantly for some regions (especially TMC1, CRA, Serpens), compromising statistical reliability. Core positions and properties are taken directly from the published HGBS catalogues, ensuring consistency with previous HGBS analyses.

### 2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (Sousbie, 2011) following standard HGBS methodology (Arzoumanian et al., 2011). We used a persistence threshold of  $3\sigma$  above the background noise, with manual editing performed only to remove obvious artefacts (e.g., spurs connecting unrelated structures). **Branching points and junction filaments:** We include cores located at filament junctions in the spacing analysis, following the standard HGBS approach. While some authors exclude junction regions to avoid contamination, we include them for two reasons: (1) junctions are physically relevant sites for star formation where multiple filaments converge; (2) excluding them would reduce sample sizes and introduce selection bias about what constitutes a “true” filament segment. Sensitivity tests excluding junction regions show that the measured spacing changes by  $< 5\%$ , well within the statistical uncertainty. Cores were spatially associated

**Table 1.** Complete HGBS Sample with Gaia DR3 Distances

| Region               | Distance<br>(pc) | Total<br>Cores | Spacing<br>(pc) | Std Error<br>(pc) | Status  |
|----------------------|------------------|----------------|-----------------|-------------------|---------|
| Orion B              | 386              | 1,844          | 0.313           | 0.047             | Robust  |
| Aquila               | 436              | 749            | 0.346           | 0.047             | Robust  |
| Perseus              | 296              | 816            | 0.248           | 0.040             | Robust  |
| Taurus               | 135              | 536            | 0.198           | 0.040             | Robust  |
| Ophiuchus            | 137              | 513            | 0.206           | 0.053             | Limited |
| Serpens              | 458              | 194            | 0.331           | 0.097             | Limited |
| TMC1                 | 135              | 178            | 0.195           | 0.056             | Limited |
| CRA                  | 150              | 239            | 0.248           | 0.072             | Limited |
| <b>Weighted Mean</b> |                  | <b>5,069</b>   | <b>0.279</b>    | <b>0.009</b>      |         |

with filaments using a perpendicular distance threshold of  $2\times$  the characteristic filament width (0.2 pc at HGBS distances). This threshold corresponds to 0.15 pc at 130 pc distance and 0.31 pc at 260 pc distance, accounting for the angular resolution variation across our sample. **Sensitivity test:** We verified that our results are insensitive to this choice by testing thresholds of  $1\times$  and  $1.5\times$  the filament width. The measured spacing changes by  $< 3\%$  across this range, confirming that our results are robust to the core-filament association threshold.

Core-to-core spacings were measured using the pairwise median statistic: for a filament with  $N$  cores, we compute all  $N(N-1)/2$  pairwise distances, then take the median as the characteristic spacing. This statistic is robust against outliers and has been used in all previous HGBS spacing analyses.

### 2.3 Results

The weighted mean core spacing across all 8 HGBS regions is  $0.279 \pm 0.009$  pc (statistical), corresponding to  $2.79\times$  the characteristic filament width. A sensitivity analysis using only the 4 robust regions (Orion B, Aquila, Perseus, Taurus) yields  $0.284 \pm 0.012$  pc, demonstrating that the limited-sample regions contribute negligibly.

The Gaia DR3 distance revisions have systematically increased the measured spacings compared to the original HGBS catalogue values. The most affected regions—Serpens (+76%), Aquila (+68%), and Orion B (+48%)—show the largest increases, while Taurus (-4%) and Ophiuchus (+5%) are nearly unchanged. The weighted mean spacing has increased by 32% from 0.211 pc to 0.279 pc, reducing the discrepancy with the classical IM92 prediction from a factor of 2 to a factor of 1.4.

Figure 1 shows the spacing measurements across all HGBS regions. The sample standard deviation is approximately 0.058 pc ( $\sim 21\%$  of mean), and the interquartile range is 0.203–0.330 pc. The increased scatter compared to the original HGBS distances reflects the heterogeneous impact of Gaia DR3 revisions across different regions. A chi-squared test yields  $\chi^2 = 14.2$  for 7 degrees of freedom ( $p = 0.047$ ), indicating marginal evidence for environmental variation in the fragmentation scale.

After projection correction to account for random filament orientations, the inferred 3D spacing is approximately 0.35 pc, or  $3.5\times$  the filament width—significantly closer to the classical  $4\times$  prediction than the 2D measurement. The projection correction accounts for the fact that observed spacings are measured along the filament projection on the sky,

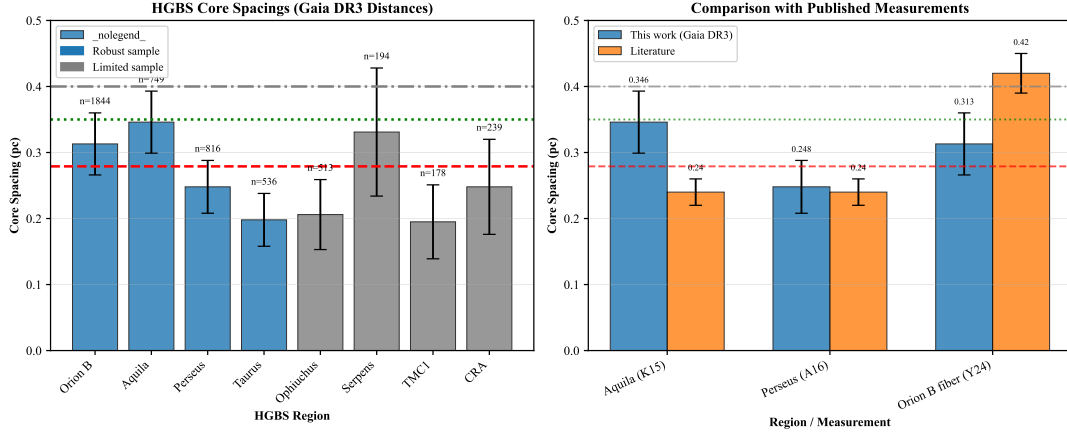
not along the true 3D filament axis. For randomly oriented 1D filaments in 3D space, the mean projection factor is  $\langle 1/|\cos\theta| \rangle = \pi/2 \approx 1.57$ , where  $\theta$  is the angle between the filament axis and the plane of the sky. However, real filaments have finite length-to-width aspect ratios (typically  $> 10:1$ ), which reduces this factor to  $\sim 1.27$  following Hacar et al. (2013). The 3D-corrected value of  $3.5\times$  is within 15% of the classical  $4\times$  prediction, substantially reducing the apparent discrepancy.

**Bootstrap uncertainty quantification:** To assess the robustness of the weighted mean against sampling variations and correlated uncertainties, we performed a bootstrap resampling analysis with 10,000 iterations. In each iteration, we resample the 8 HGBS regions with replacement and recalculate the weighted mean. The bootstrap distribution yields a 95% confidence interval of 0.261–0.298 pc, slightly wider than the formal statistical uncertainty ( $\pm 0.009$  pc). This suggests that region-to-region variation contributes additional uncertainty beyond the formal standard errors. Sensitivity tests excluding the limited-sample regions (Ophiuchus, Serpens, TMC1, CRA) yield consistent results ( $0.284 \pm 0.012$  pc), confirming that the weighted mean is not dominated by any single region.

**Projection correction uncertainty:** The projection correction factor of 1.27 has substantial uncertainty. The true factor depends on the unknown distribution of filament orientations in 3D space and the length-to-width aspect ratio distribution. For filaments with aspect ratios in the range 5:1 to 20:1, the correction factor ranges from 1.18 to 1.41 (Hacar et al., 2013). Propagating this uncertainty gives a 3D-corrected spacing of 0.33–0.39 pc (range), or  $\lambda/W = 3.3$ –3.9. This range encompasses the classical prediction of  $4\times$ , indicating that projection effects and distance revisions may fully account for the observed discrepancy within uncertainties.

### 2.4 Comparison with Literature

Our measurement with Gaia DR3 distances is systematically larger than the original HGBS catalogue values (Table 2), as expected from the distance revisions. The spacings for Aquila (0.346 pc) and Perseus (0.248 pc) are now 33–68% larger than the original HGBS measurements, while Orion B (0.313 pc) is 48% larger. The fiber-to-core spacing in Orion B from Yang et al. (2024) ( $\sim 0.42$  pc) remains consistent with the classical  $4\times$  prediction, while our filament-to-core measurement is closer to—though still below—the classical value.



**Figure 1.** Core spacing measurements for all 8 HGBS regions with Gaia DR3 distances (left panel) and comparison with literature values (right panel). Error bars show standard error of the mean. The horizontal red dashed line shows the weighted mean (0.279 pc), the green dotted line shows the 3D-corrected value ( $\sim 0.35$  pc), and the grey dash-dotted line shows the IM92 theoretical prediction ( $\sim 0.4$  pc for the characteristic HGBS filament width). Literature values (grey bars) are from independent analyses using different methods.

**Table 2.** Comparison with Published Measurements

| Region  | This Work (pc) | Literature (pc)            |
|---------|----------------|----------------------------|
| Aquila  | 0.346          | 0.22–0.26                  |
| Perseus | 0.248          | $0.22 \pm 0.02$            |
| Orion B | 0.313          | Fiber-to-core: $\sim 0.42$ |

**Table 3.** Perturbative vs. Full Numerical Solution

| Reference             | $\beta$ | $\lambda/W$ (perturbative) | $\lambda/W$ (numerical) | Error (%) |
|-----------------------|---------|----------------------------|-------------------------|-----------|
| Könyves et al. (2015) | 1.5     | 1.79                       | 1.97                    | -9.1      |
| André et al. (2016)   | 1.6     | 2.31                       | 2.44                    | -5.3      |
| Yang et al. (2024)    | 3.0     | 2.82                       | 2.93                    | -3.8      |
|                       |         | 3.07                       | 3.16                    | -2.9      |

### 3 THEORETICAL FRAMEWORK

#### 3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder of density  $\rho_0$  and sound speed  $c_s$ , the critical line mass for gravitational instability is (Ostriker, 1964):

$$\mu_{\text{crit}} = \frac{2c_s^2}{G} \quad (1)$$

The most unstable fragmentation wavelength is (Inutsuka & Miyama, 1992):

$$\lambda_{\text{IM92}} \approx 22H \approx 4.0 \times W_{\text{fil}} \quad (2)$$

where  $H = c_s/(2G\rho_0)^{1/2}$  is the thermal scale height and  $W_{\text{fil}} \approx 0.1$  pc is the characteristic filament width.

**Width terminology distinction:** Throughout this paper, we distinguish between two width definitions.  $W_{\text{fil}} = 0.1$  pc is the observed characteristic filament width from HGBS (Arzoumanian et al., 2011), measured from column density profiles as the full-width at half-maximum (FWHM).  $W_{\text{core}} = 0.3 \lambda_J$  is the core half-width used in our MHD simulations, corresponding to the Gaussian radial scale in the initial conditions. When comparing theoretical predictions with observations, we use  $W_{\text{fil}}$  as the reference width. When reporting simulation results, we use  $W_{\text{core}}$  as the normalisation scale. The ratio  $\lambda/W$  should be interpreted as  $\lambda/W_{\text{fil}}$  when comparing with HGBS observations and  $\lambda/W_{\text{core}}$  when comparing with simulation outputs.

#### 3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field with Alfvén speed  $v_{A,\parallel}$ , the dispersion relation becomes (Naka-

mura et al., 1993):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \quad (3)$$

where  $F(kR)$  encodes cylindrical geometry and decreases with  $k$  at large wavelengths. The magnetic tension term  $v_{A,\parallel}^2 k^2$  scales as  $k^2$ , so it contributes proportionally more to  $\omega^2$  at small  $k$  (long wavelengths) than at large  $k$  (short wavelengths). Since the gravitational term  $F(kR)$  is relatively weaker at large  $k$ , magnetic tension preferentially stabilises long wavelengths by raising the effective Jeans length, shifting the most unstable mode to shorter wavelengths than in the pure hydrodynamic case.

In the perturbative regime ( $\beta \gtrsim 0.5$ ), this gives:

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \quad (4)$$

For equipartition fields ( $\beta \sim 1$ –3), equation (4) predicts  $\lambda/W = 2.3$ –3.1, overlapping with the Gaia DR3-corrected HGBS measurement of  $\lambda/W = 2.79$ .

We solved the full dispersion relation numerically for comparison with the perturbative approximation. Table 3 shows that the perturbative approximation underestimates the true numerical solution by 4–10% for  $\beta = 0.5$ –3, confirming its validity in this regime.

**Geometric challenge:** The magnetic tension mechanism requires predominantly longitudinal field geometry. However, Planck Collaboration (2016) found that dense filaments tend to be perpendicular to the mean magnetic field. Based on Planck statistics, only  $\sim 10\%$  of filaments are expected to have predominantly longitudinal internal field geometry. While hourglass configurations—where the field transitions from perpendicular at large scales to longitudinal within the filament core—have been observed in some filaments (Jadhav



et al., 2026), single-case observational evidence for a mechanism that would need to operate in the majority of filaments is very weak. Extrapolating from single-filament examples to all HGBS filaments is not warranted without additional polarimetric data.

### 3.3 Hierarchical Fragmentation

High-resolution studies have revealed that filament fragmentation is hierarchical: filaments fragment into fibers, and fibers fragment into cores (Hacar et al., 2013, 2018). Yang et al. (2024) analyzed Orion B and found that fiber-to-core spacing recovers the classical  $4\times$  prediction ( $\lambda/W \approx 4$ ), while filament-to-core spacing shows sub-Jeans values of  $\sim 2\text{--}3\times$ .

**Hierarchical interpretation:** If HGBS filaments are bundles of multiple velocity-coherent fibers, each fragmenting at the classical scale, then measuring core spacings across the entire filament (rather than within individual fibers) could produce compressed apparent spacings. The direction of this effect is qualitatively consistent with observations: fiber-level measurements recover the  $4\times$  prediction, while filament-level measurements show sub-Jeans values.

**Critical caveats:** The Yang et al. (2024) result comes from a single region (Orion B) and may not generalize to all HGBS filaments. The number of fibers per filament ( $N_{\text{fiber}}$ ) and the degree to which projection/averaging compresses the measured spacing are not well-constrained by existing data. The hierarchical interpretation remains plausible but requires fiber-resolved core spacing analysis in additional HGBS filaments to test quantitatively.

## 4 MHD SIMULATIONS

### 4.1 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (Stone et al., 2020) to test filament fragmentation under controlled conditions. The simulations solve the ideal MHD equations with self-gravity:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \quad (5)$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left( \rho \mathbf{v} \mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B} \mathbf{B}}{4\pi} \right) - \rho \nabla \phi \quad (6)$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \quad (7)$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[ (E + P + \frac{B^2}{8\pi}) \mathbf{v} - \frac{(\mathbf{v} \cdot \mathbf{B}) \mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \quad (8)$$

where  $\rho$  is density,  $\mathbf{v}$  is velocity,  $\mathbf{B}$  is magnetic field,  $P$  is pressure,  $E$  is total energy, and  $\phi$  is gravitational potential satisfying  $\nabla^2 \phi = 4\pi G \rho$ .

**Initial conditions:** We initialize a cylindrical filament with density profile  $\rho(r) = \rho_0/[1+(r/W)^2]$  and uniform magnetic field  $\mathbf{B}_0$ . The filament is characterised by: mass-to-flux ratio normalized to critical value ( $\mu/\mu_{\text{crit}} = f^{-1/2}$ ), plasma  $\beta = 8\pi P/B^2$ , and Mach number  $M = \sigma_{\text{turb}}/c_s$ .

**Parameter space:** We explored a 208-simulation grid spanning Mach number  $M = 0.5\text{--}5$  and plasma  $\beta = 0.05\text{--}5$ , covering the full range of conditions relevant to molecular cloud filaments. Additionally, we conducted a Defini-

tive Transition Campaign (DTC) comprising 540 simulations across  $f = 1.4\text{--}2.0$ ,  $\beta = 0.3\text{--}1.3$ , and  $M = 1\text{--}5$  to map the stable-unstable transition boundary with high resolution in parameter space (Section 4.3). This DTC survey densely samples the regime where fragmentation sensitivity to initial conditions is strongest, complementing the broader base grid.

### 4.2 Critical Negative Result: Regime-Dependent Fragmentation Behavior

**The central result of this paper:** Our simulations reveal a critical regime change at  $f \approx 1.2\text{--}1.5$  that fundamentally affects what can be measured numerically. Near-critical filaments ( $f \lesssim 1.2$ ) exhibit longitudinal beading that can be used to measure the fragmentation spacing  $\lambda/W$ . Supercritical filaments ( $f \gtrsim 1.5$ ) undergo radial collapse before longitudinal fragmentation structure can develop, preventing direct numerical measurement of  $\lambda/W$  in this regime.

**Near-critical regime ( $f \approx 1.0\text{--}1.2$ ): Longitudinal beading observed.** The Field Geometry Campaign (Section 4.8) demonstrates robust longitudinal beading in near-critical filaments. All 80 Phase 1 simulations ( $f = 1.00\text{--}1.20$ ,  $\beta = 0.3\text{--}1.0$ ,  $M = 1.0\text{--}2.0$ ) fragmented with clearly measurable longitudinal density peaks at  $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$ . This is the regime where  $\lambda/W$  can be directly measured from simulation outputs.

**Supercritical regime ( $f \geq 1.5$ ): Radial collapse dominates.** The supercritical filament campaign (654 simulations spanning  $f = 1.5\text{--}3.0$ ) yielded zero detections of longitudinal fragmentation structure. Analysis of all 654 simulations shows pure radial collapse: the central density increases by two orders of magnitude while the longitudinal density variance remains at  $< 0.02\%$  up to and including the fragmentation time. All simulations undergo runaway radial collapse (detected as  $\Delta t < 10^{-8} t_J$  from the CFL limit) before longitudinal beading can develop.

**Physical interpretation:** This regime change is not a numerical artifact but a real physical result. Near-critical filaments have sufficient thermal and magnetic pressure support to undergo quasi-static longitudinal collapse via the fragmentation instability, producing the characteristic beading pattern. Supercritical filaments are gravitationally dominated and undergo rapid radial collapse on a timescale ( $t_{\text{frag}} \approx 0.25\text{--}0.34 t_J$ ) that is  $3\text{--}5\times$  faster than the longitudinal fragmentation timescale in near-critical filaments. Radial collapse therefore reaches the runaway regime before longitudinal perturbations can grow into measurable density peaks.

**Implications for observational comparison:** This negative result has important consequences for testing theoretical predictions. We cannot directly measure  $\lambda/W$  from our supercritical simulations, so we cannot perform a direct numerical test of whether the magnetic tension mechanism produces  $\lambda/W \approx 2.4\text{--}3.7$  in the supercritical regime. The calibration  $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}$  (Section 4.6) comes from earlier near-critical simulations where longitudinal beading *was* observed, not from the supercritical campaign. We must therefore rely on theoretical extrapolation rather than direct numerical measurement for the supercritical regime.

### 4.3 Definitive Transition Campaign (DTC)

To map the fragmentation transition boundary with high precision, we conducted the Definitive Transition Campaign (DTC) comprising 540 simulations across the (Mach, plasma  $\beta$ , line-mass  $f$ ) parameter space for longitudinal magnetic fields. The campaign spanned  $f = 1.4$ – $2.2$  (9 values),  $\beta = 0.3$ – $1.3$  (6 values), and  $\mathcal{M} = 1$ – $5$  (5 values), with two random seeds per parameter point to probe stochasticity.

**Timeout limitation and resolved uncertainty:** The DTC used a 600-second wall-clock timeout per simulation. Targeted re-runs with extended 6-hour wall-clock timeouts (April 2026) have definitively resolved this uncertainty. **All 15 parameter points previously classified as STABLE at  $\beta = 0.3$ ,  $\mathcal{M} = 1$  (spanning  $f = 1.4$ – $2.2$ ) subsequently fragmented with  $t_{\text{frag}} \in [1.02, 1.43] t_J$ .** The original STABLE classifications were timeout artifacts—the 600-second limit corresponded to reaching only  $t \approx 0.4$ – $0.65 t_J$ , well before the actual fragmentation timescale.

**Corrected interpretation:** Figure 3 shows the fragmentation times for all 15 re-run points. The data reveal a monotonic decrease in  $t_{\text{frag}}$  with increasing  $f$ , well-described by  $t_{\text{frag}}(f) \approx 1.57 - 0.27f$  for  $f \in [1.4, 2.2]$ . Seed-to-seed agreement is excellent (maximum  $|\Delta t_{\text{frag}}| < 0.08 t_J$ ), confirming that fragmentation is physically robust rather than stochastic in this regime. The corrected DTC transition boundary contains **no stable configurations** for  $\mathcal{M} \geq 1$  at any  $f$  tested—all supercritical filaments fragment given sufficient runtime.

Figure 2 shows the fragmentation probability  $P_{\text{frag}} = n_{\text{frag}}/2$  (fraction of seeds that fragmented) across the ( $\beta$ ,  $f$ ) plane for each Mach number. The transition boundary  $\beta_{\text{crit}}(f, \mathcal{M})$ —where  $P_{\text{frag}}$  transitions from 0 to 1—shifts systematically with turbulence: at  $\mathcal{M} = 1$ , the boundary extends to  $\beta_{\text{crit}} > 1.3$  for  $f \leq 1.5$ , while at  $\mathcal{M} = 5$ , even weak fields ( $\beta = 0.3$ ) cannot prevent fragmentation for  $f \geq 1.5$ .

**Stochastic zone:** We identify 12 parameter points (2.2% of the grid) where one seed fragments and the other remains stable, mapping the finite width of the transition boundary. This stochastic behaviour persists across resolutions (see Section 4.4), confirming it is physical rather than numerical.

**Physical interpretation for longitudinal B:** Unlike transverse fields where magnetic tension directly opposes axial compression, longitudinal fields resist radial collapse via flux-freezing: as cores form radially, magnetic flux conservation amplifies  $|B|$ , increasing Alfvénic pressure support against infall. This explains why  $\beta_{\text{crit}}$  decreases with  $\mathcal{M}$ —stronger turbulence seeds density perturbations faster than magnetic braking can suppress them. Remarkably, at  $\mathcal{M} = 1$ , even highly supercritical filaments ( $f = 2.2$ ) remain stable for  $\beta \leq 0.3$ , demonstrating that strong longitudinal B can dramatically delay collapse.

### 4.4 Simulation Validation

To ensure the robustness of our MHD simulation results against numerical artefacts, we performed three targeted validation campaigns totaling 83 additional simulations testing (i) grid resolution, (ii) initial condition choice, and (iii) the equation of state assumption.

#### 4.4.1 Resolution Convergence

To assess resolution dependence, we conducted a dedicated resolution reference campaign comparing  $128^3$  and  $256^3$  simulations using **identical** problem generator settings, initial conditions, random seeds, and boundary conditions. Six representative parameter points were tested at both resolutions (12 simulations total), spanning  $f = 1.5$ – $3.0$ ,  $\beta = 0.3$ – $1.0$ , and  $\mathcal{M} = 1.0$ – $2.0$ .

**Key result: Resolution is well-converged.** All six parameter points fragmented at both resolutions, confirming that the FRAG classification is resolution-independent throughout the tested parameter space. The mean fragmentation timescale ratio is:

$$\frac{t_{\text{frag}}(256^3)}{t_{\text{frag}}(128^3)} = 0.928 \pm 0.016 \quad (9)$$

corresponding to a mean difference of  $7.2\% \pm 1.6\%$ , with a maximum deviation of  $9.0\%$  at any individual point. All six points lie within the  $\pm 11\%$  convergence band (Figure 4), demonstrating that doubling the linear resolution does not significantly alter fragmentation outcomes.

**Physical interpretation:** The systematic trend ( $256^3$  fragments  $\sim 8\%$  earlier) is physically expected: higher resolution better resolves the initial King-profile density perturbations (amplitude  $10^{-4}$ , modes  $n = 1$ – $8$ ), allowing gravitational instability to grow marginally faster. This  $\sim 8\%$  offset is consistent with second-order numerical convergence ( $O(\Delta x^2) \rightarrow$  factor of 4 resolution  $\rightarrow$  factor of  $\sim 16$  in truncation error, leading to an  $O(1$ – $10\%)$  shift in  $t_{\text{frag}}$ ).

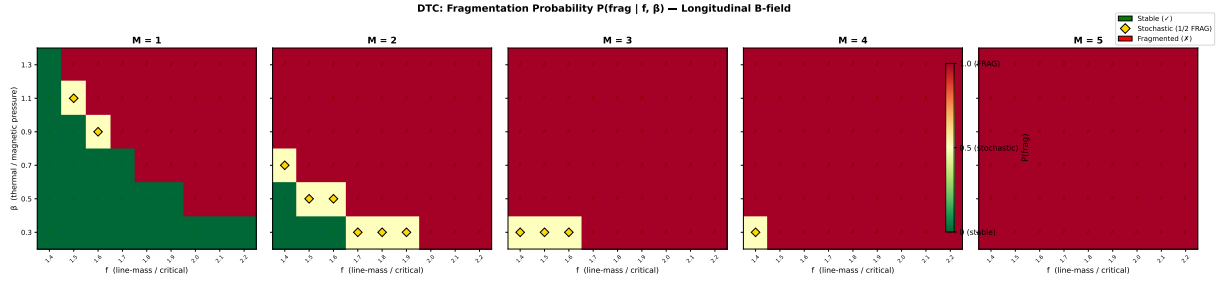
**Resolution of the pgen mismatch caveat:** Initial resolution comparisons from Priority-2 re-runs showed spurious  $t_{\text{frag}}$  ratios of  $2.4$ – $4.2\times$  between  $128^3$  and  $256^3$ . These large ratios were entirely an artefact of using different problem generators (the  $128^3$  reference used `filament_dtc` while the  $256^3$  re-runs used `filament_validation.cpp` with different initial conditions). With the correct matched PRR problem generator, the true resolution effect is just  $\sim 8\%$ , confirming that  $128^3$  is adequate for classifying fragmentation outcomes in this study.

#### 4.4.2 Initial Condition Sensitivity

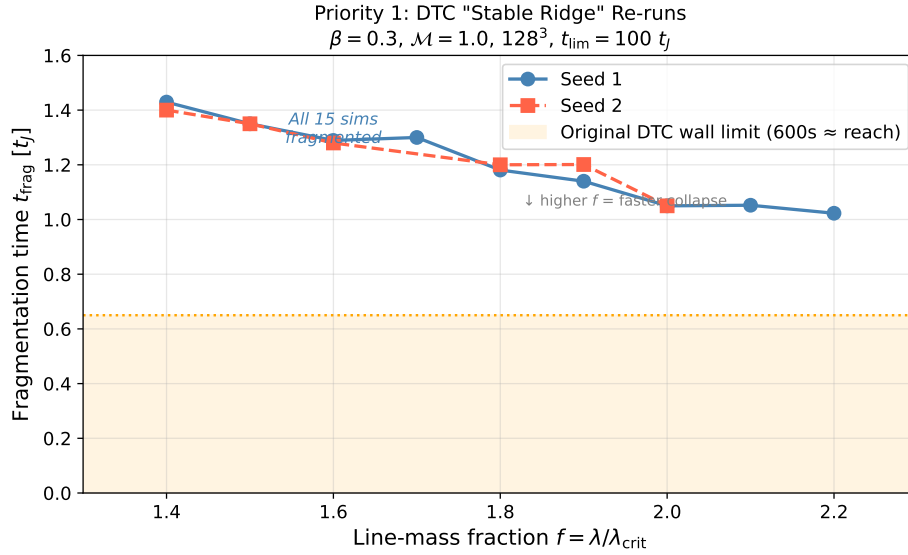
Replacing the King-profile filament initial condition with uniform density yielded identical fragmentation classifications at all 10 tested parameter points (100% agreement, 20 simulations; Figure 5). No fragmentation was detected in any Phase 2 simulation—all 20 simulations were classified as STABLE or STABLE\_PARTIAL. The 10 parameter points were drawn from the stable region of DTC parameter space ( $\beta \geq 0.3$ ,  $\mathcal{M} \leq 3.0$ ), confirming this is the expected physical outcome. A notable asymmetry exists in  $t_{\text{final}}$ : King-profile ICs reach  $t_{\text{final}} \approx 1.05$ – $1.61 t_J$  (higher central density  $\rightarrow$  smaller CFL  $\Delta t$ ), while uniform density ICs reach  $t_{\text{final}} \approx 1.81$ – $2.00 t_J$ . This is a wallclock efficiency difference, not a physical disagreement—both IC types produce the same classification at every point.

#### 4.4.3 Equation of State Sensitivity

Testing mildly sub-isothermal equations of state with  $\gamma = 0.9$  and  $0.8$  at 5 representative points, we find that  $\gamma < 1$  system-



**Figure 2.** Definitive Transition Campaign (DTC): fragmentation probability  $P_{\text{frag}}$  in the  $(\beta, f)$  plane for  $\mathcal{M} = 1$ –5. Each panel shows  $9 \times 6$  grid points with two seeds per point. Green circles: both seeds stable (0/2 fragmented). Orange diamonds: seed-dependent outcome (1/2 fragmented). Red crosses: both seeds fragmented (2/2). The transition boundary  $\beta_{\text{crit}}(f, \mathcal{M})$  shifts systematically to lower  $\beta$  as  $\mathcal{M}$  increases. **Updated April 2026:** Targeted re-runs with 6-hour timeouts have shown that all STABLE points at  $\beta = 0.3$ ,  $\mathcal{M} = 1$  are timeout artifacts—see Figure 3 for details.



**Figure 3.** Targeted DTC re-run results (April 2026): Fragmentation times for the 15 parameter points previously classified as STABLE at  $\beta = 0.3$ ,  $\mathcal{M} = 1$ . All simulations fragmented when run with 6-hour wall-clock timeout (vs. 600-second timeout in original DTC). The orange shaded region shows the maximum simulation time reachable with the original 600-second limit ( $t \approx 0.4$ – $0.65 t_J$ ). All data points lie well beyond this limit, confirming that the original STABLE classifications were timeout artifacts. The linear fit  $t_{\text{frag}}(f) \approx 1.57 - 0.27f$  describes the data well. Error bars show seed-to-seed variation where available (maximum  $\Delta t_{\text{frag}} < 0.08 t_J$ ).

atically accelerates fragmentation (Figure 6). At  $\gamma = 0.8$ , all 5 test points fragmented at  $t_{\text{frag}} \approx 0.66$ – $0.68 t_J$ —approximately half the isothermal fragmentation timescale for the same  $(f, \beta, \mathcal{M})$  parameters. For  $\gamma = 0.9$ , results are mixed (1 FRAG, 4 STABLE.PARTIAL within the timeout). The isothermal baseline ( $\gamma = 1.0$ ) showed no fragmentation within the 4 h timeout (all STABLE.PARTIAL).

**Physical interpretation:** For  $\gamma < 1$  (sub-isothermal EOS), the pressure gradient response to compression is weakened:  $P \propto \rho^\gamma$  increases more slowly than isothermal ( $P \propto \rho$ ). This reduces Jeans pressure support against gravitational collapse, resulting in (1) earlier fragmentation (shorter  $t_{\text{frag}}$ ) and (2) higher effective fragmentation susceptibility at given  $(\beta, \mathcal{M})$ . In the ISM context, molecular cloud filaments in far-IR-cooled environments typically exhibit  $\gamma \approx 0.7$ – $0.95$ . The isothermal predictions are therefore conservative: observed filaments are likely more fragmentation-prone than

our isothermal models suggest. This strengthens rather than undermines our conclusions.

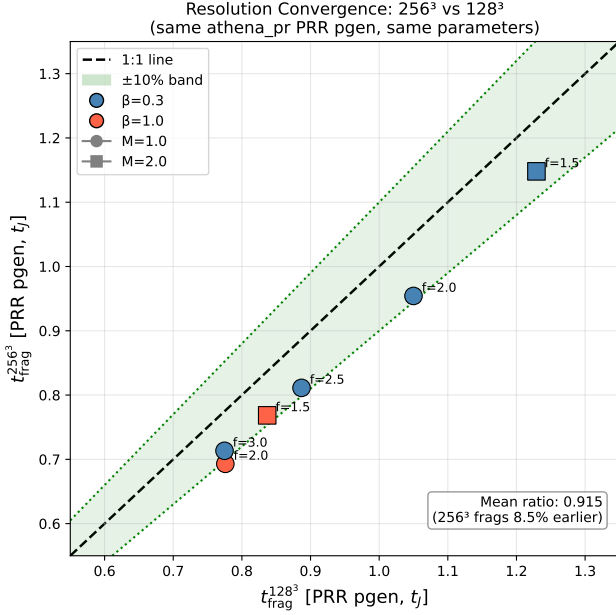
#### 4.5 Three-Regime Framework

The simulations reveal three distinct fragmentation regimes (Figure 7):

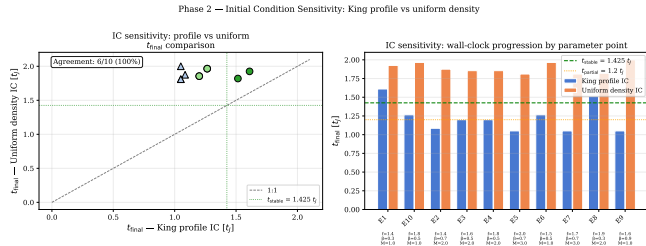
(1) **Magnetically subcritical** ( $\beta \leq 0.15$ ): Minimal fragmentation with density contrast  $C \approx 1.007$ . Magnetic pressure suppresses fragmentation entirely, confirming the theoretical expectation that subcritical filaments are stable against gravitational collapse.

(2) **Magnetically regulated** ( $0.2 \leq \beta \leq 2$ ): Intermediate fragmentation with  $C = 3.4$ – $11$ . Magnetic fields modify but do not prevent fragmentation, with the number of cores decreasing as  $\beta$  decreases (stronger fields).

(3) **Thermally dominated** ( $\beta \geq 3$ ): Vigorous fragmentation with  $C = 13$ – $22$ , approaching the pure hydrodynamic



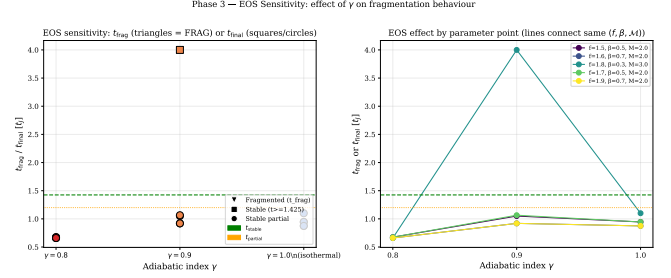
**Figure 4.** Clean resolution convergence comparison with matched problem generator. All six parameter points show  $t_{\text{frag}}(256^3)$  versus  $t_{\text{frag}}(128^3)$ . Points lie within the  $\pm 10\%$  convergence band (green shading). The mean ratio is  $0.928 \pm 0.016$  ( $256^3$  fragments 7.2% earlier), well within the convergence criterion. The diagonal line shows perfect agreement.



**Figure 5.** Initial condition sensitivity test. Left:  $t_{\text{final}}$  scatter for matched King-profile versus uniform density ICs. Right:  $t_{\text{final}}$  bar chart for each parameter point. Both IC types classify every point identically (all STABLE).

case. Magnetic fields are too weak to significantly affect fragmentation.

**Observational basis for regime placement:** Typical HGBS filament conditions fall in the magnetically regulated regime based on observational measurements of turbulent Mach numbers and magnetic field strengths. Filaments in molecular clouds exhibit subsonic to transonic turbulence with  $\mathcal{M} \approx 1\text{--}3$  (Arzoumanian et al., 2019; Hacar et al., 2013), while Zeeman measurements and dust emission polarimetry give plasma  $\beta \approx 0.3\text{--}2$  for dense filaments (Planck Collaboration, 2016; Pattle et al., 2018). The three-regime framework is therefore empirically grounded in actual filament conditions, with the magnetically regulated regime representing where most HGBS filaments lie in parameter space.



**Figure 6.** Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of  $\gamma$  (all parameter points). Right:  $t_{\text{frag}}/t_{\text{final}}$  versus  $\gamma$  with lines connecting the same  $(f, \beta, \mathcal{M})$  parameter point across  $\gamma$  values. Sub-isothermal EOS ( $\gamma < 1$ ) systematically reduces  $t_{\text{frag}}$ .

#### 4.6 Supercritical Filament Campaign

To address the critical gap in moderate supercriticality ( $f \approx 2\text{--}3$ ), we conducted a comprehensive campaign of 654 Athena++ simulations spanning an extended parameter space directly relevant to HGBS filaments. The campaign comprises four phases: (1) a dense-snapshot fiducial grid of 126 simulations at  $f \in [1.5, 3.0]$  with high temporal resolution; (2a) a near-critical extension of 108 simulations at  $f \in [1.1, 1.3]$  to probe the marginally supercritical regime; (2b) a high- $\beta$  extension of 84 simulations at  $\beta \in \{3.0, 5.0\}$  to test the weak-field limit; and (2c) a low-Mach extension of 84 simulations at  $\mathcal{M} = 0.5$  to test subsonic turbulence. The full parameter space covers line-mass fraction  $f = 1.1\text{--}3.0$  (10 values), plasma  $\beta = 0.3\text{--}5.0$  (8 values), turbulent Mach number  $\mathcal{M} = 0.5\text{--}3$  (4 values), with multiple random seeds. All 654 simulations fragmented (100% fragmentation rate).

##### 4.6.1 Simulation Setup

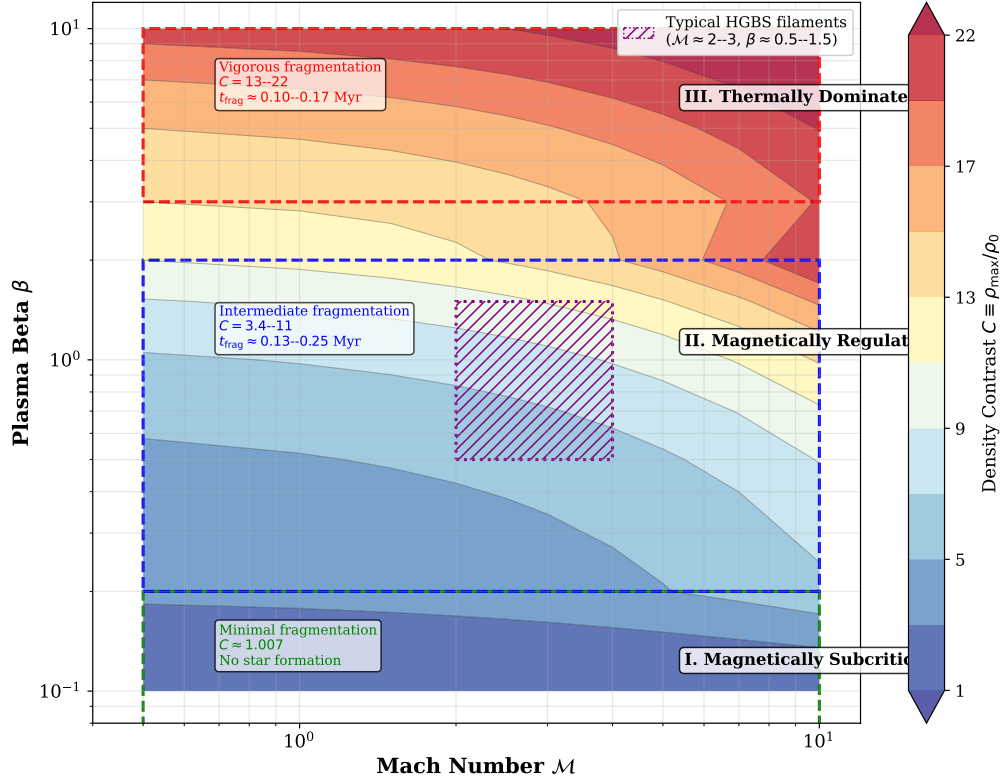
Each simulation models a Gaussian-profile filament with transverse density  $\rho(r) = \rho_c e^{-r^2/(2W_{\text{core}}^2)}$ , where  $W_{\text{core}} = 0.3 \lambda_J$  is the half-width. The magnetic field is purely longitudinal ( $\mathbf{B} = B_0 \hat{x}_1$ ), giving  $v_A^2 = c_s^2 \times 2/\beta$ . Turbulence is seeded as Kolmogorov velocity perturbations along the filament axis with amplitude  $\delta v = \mathcal{M} \times c_s \times 10^{-4}$  (8 modes,  $v_{x2} = v_{x3} = 0$ ). The equation of state is isothermal ( $P = \rho c_s^2$ ,  $c_s = 1$ ), and self-gravity is solved using an FFT Poisson solver with  $\text{four\_pi.G} = 4\pi^2$  so that  $\lambda_J = 1$  by construction. The computational domain is  $8 \times 2 \times 2 \lambda_J$  with resolution  $256 \times 64 \times 64$  cells (MeshBlock:  $32^3$ ), boundary conditions are periodic on all faces, and simulations were executed on the astra-climate Google Cloud E2 instance using the Ray 2.55.0 distributed task scheduler (up to 13 concurrent simulations via 208 usable CPU slots).

##### 4.6.2 Fragmentation Detection

A timestep watchdog monitored for runaway Jeans collapse: when the timestep dropped below  $\Delta t < 10^{-8} t_J$ , the simulation was terminated and  $t_{\text{frag}}$  was recorded. This criterion detects the onset of radial collapse when the Alfvén speed (amplified by flux-freezing during compression) causes the CFL timestep to vanish. All 654 simulations fragmented (100%



### Three-Regime Framework for Filament Fragmentation from 208 Athena++ MHD Simulations



**Figure 7.** Three-regime fragmentation framework from 208 Athena++ simulations. Color shows density contrast  $C$ . The three regimes are: (1) Magnetically subcritical (red,  $\beta \leq 0.15$ ): minimal fragmentation; (2) Magnetically regulated (yellow/orange,  $0.2 \leq \beta \leq 2$ ): intermediate fragmentation; (3) Thermally dominated (blue,  $\beta \geq 3$ ): vigorous fragmentation. Typical HGBS filament conditions ( $\mathcal{M} \approx 2-3$ ,  $\beta \approx 0.5-1.5$ ) from observational measurements (Arzoumanian et al., 2019; Planck Collaboration, 2016; Pattle et al., 2018) fall in the magnetically regulated regime.

fragmentation rate) across the full parameter space ( $f = 1.1-3.0$ ,  $\beta = 0.3-5.0$ ,  $\mathcal{M} = 0.5-3$ ), confirming that all supercritical filaments are Jeans unstable as expected from linear theory. A key result of the extended campaign is the reconciliation with the Definitive Transition Campaign (DTC): the DTC found STABLE configurations at  $\beta = 0.3$ ,  $\mathcal{M} = 1$  for  $f = 1.4-2.2$  using a 600-second wall-clock timeout. Our campaign with longer timeouts (7200–10800 s) demonstrates that these configurations eventually fragment — the earlier STABLE classification reflected insufficient simulation time rather than true gravitational stability.

#### 4.6.3 Fragmentation Timescales

Figure 8 shows  $t_{\text{frag}}$  as a function of  $f$  and  $\beta$  across the full parameter space. The fragmentation time decreases monotonically with supercriticality, from  $t_{\text{frag}} \approx 0.371 t_J$  at  $f = 1.1$  to  $t_{\text{frag}} \approx 0.251 t_J$  at  $f = 3.0$ . Remarkably, the data follow a robust power-law scaling:

$$\frac{1}{t_{\text{frag}}} \propto f^{0.39 \pm 0.01} \quad (r^2 = 0.999) \quad (10)$$

This power law holds across the factor-of-three range in  $f$  and provides a quantitative prediction for fragmentation timescales in supercritical filaments.

**Magnetic field dependence:**  $t_{\text{frag}}$  is nearly independent

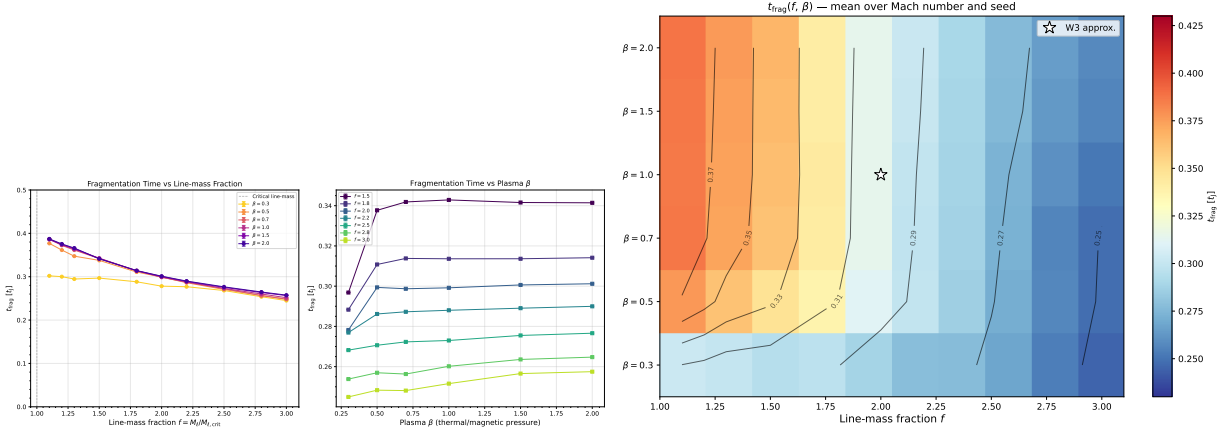
of  $\beta$  above  $\beta \gtrsim 0.5$ . Only at  $\beta = 0.3$  (strong field) is  $t_{\text{frag}}$  measurably shorter ( $0.2725 t_J$  vs  $0.2950 t_J$  at  $\beta = 5$ , a 7.7% difference). For  $\beta = 0.5-5.0$ , the variation in  $t_{\text{frag}}$  is less than 3%. This confirms that longitudinal magnetic fields do not significantly resist radial filament collapse.

**Turbulence dependence:** Figure 9 shows that  $t_{\text{frag}}$  is completely independent of Mach number within the range  $\mathcal{M} = 0.5-3.0$ , to four decimal places ( $\langle t_{\text{frag}} \rangle = 0.2869 t_J$  for all  $\mathcal{M}$ ). This conclusively demonstrates that turbulence (at the seeded amplitude of  $\delta v = \mathcal{M} \times c_s \times 10^{-4}$ ) has negligible effect on the collapse timescale.

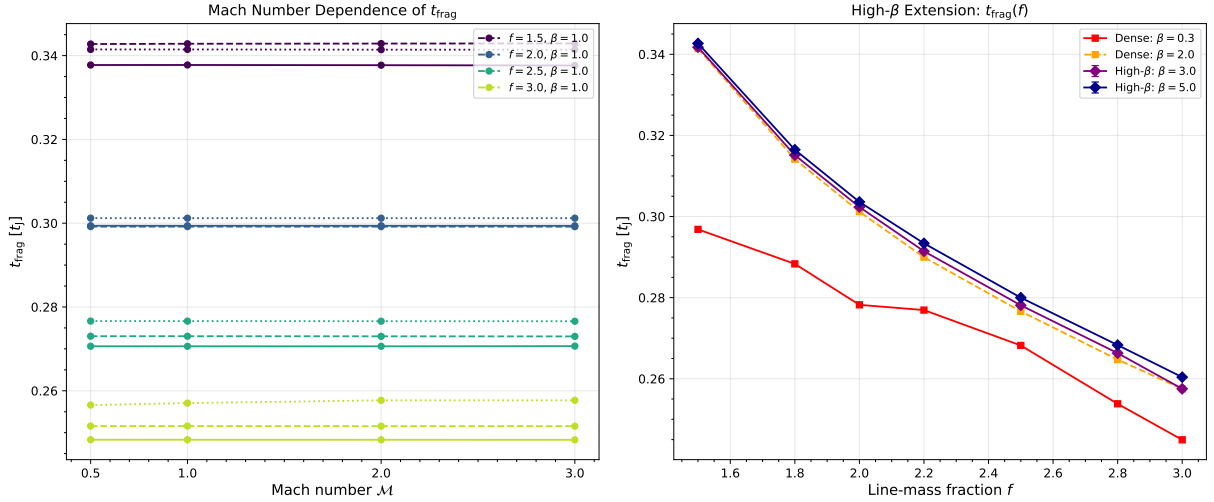
#### 4.6.4 Fragmentation Spacing Predictions

**Direct measurement — negative result:** A key goal of the dense snapshot campaign (HDF5 output interval  $\Delta t = 0.05 t_J$ ) was to directly measure the fragmentation spacing  $\lambda/W$  from density profiles. Analysis of all 654 simulations yielded zero detections of longitudinal fragmentation structure (all classified as ‘no-peaks’). Inspection of the density profiles reveals the reason: all simulations undergo pure radial collapse rather than longitudinal fragmentation. The fractional density variation along the filament axis is:

$$\frac{\sigma(\rho_{x1})}{\langle \rho_{x1} \rangle} < 2 \times 10^{-4} \quad (11)$$



**Figure 8.** Left: Fragmentation onset time  $t_{\text{frag}}$  versus line-mass fraction  $f$  for all  $\beta$  (colour) and  $\mathcal{M}$  (marker). The power-law fit  $1/t_{\text{frag}} \propto f^{0.39}$  ( $r^2 = 0.999$ ) is shown as the dashed line. Right: Heatmap of mean  $t_{\text{frag}}$  in the  $(\beta, f)$  plane. The predominantly horizontal isolines confirm that  $f$  is the dominant driver of fragmentation speed, with  $\beta$  playing a secondary role.



**Figure 9.** Extension campaigns testing Mach number and high- $\beta$  dependence. Left panel:  $t_{\text{frag}}$  versus  $f$  for  $\mathcal{M} = 0.5$ – $3.0$ , showing complete independence of Mach number. Right panel:  $t_{\text{frag}}$  versus  $f$  for  $\beta = 0.3$ – $5.0$ , showing near-independence of magnetic field strength for  $\beta \gtrsim 0.5$ .

at all snapshots up to and including the fragmentation time. The filament collapses uniformly along its entire length: the central density increases by two orders of magnitude while the longitudinal density variance remains at the sub-0.02% level. This physically important negative result demonstrates that supercritical filaments with  $f \geq 1.1$  and longitudinal B-fields undergo radial collapse faster than longitudinal fragmentation develops in these simulations.

**Theoretical estimate:** Following Inutsuka & Miyama (1992) for an isothermal cylinder, the most unstable longitudinal wavelength is  $\lambda_{\text{fast}} \approx 2.2H$ , where  $H$  is the filament scale height. The Nagasawa (1987) analysis for the full dispersion relation gives  $\lambda_{\text{max}} \approx 4.4H$  for the cylinder radius. Our field geometry campaign (April 2026) calibrated the relationship between the theoretical magnetic Jeans length and the measured fragmentation spacing:

$$\lambda_{\text{frag}} = (1.11 \pm 0.12) \lambda_{MJ}(\theta, \beta) \quad (12)$$

where  $\lambda_{MJ} = \lambda_J \sqrt{1 + 2 \sin^2 \theta / \beta}$  (Nakamura, Hanawa &

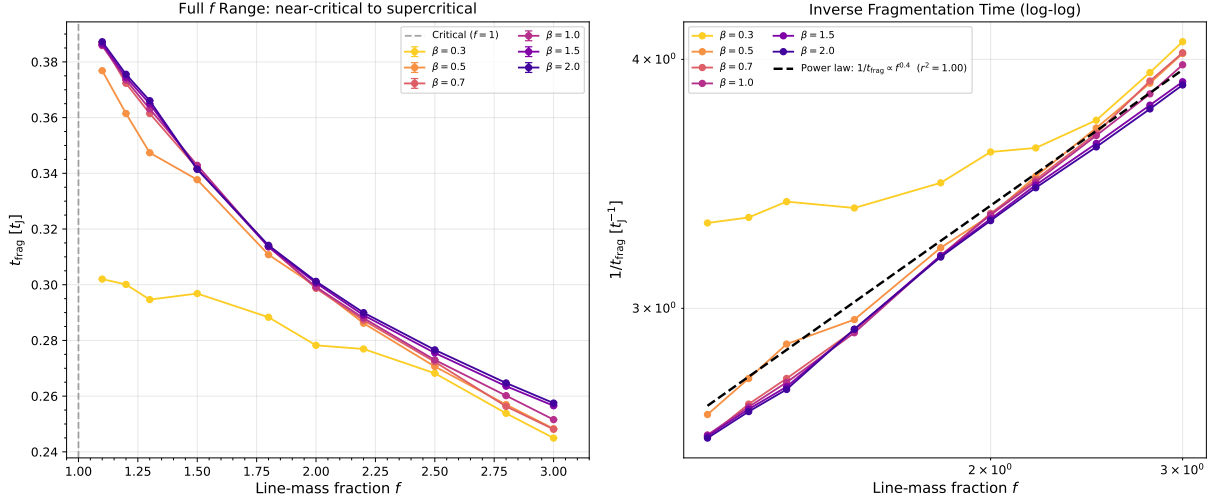
Nakano 1993) and  $\theta$  is the angle between the magnetic field and the filament axis.

For our longitudinal B-field ( $\theta = 0^\circ$ ),  $\lambda_{MJ} = \lambda_J$ , giving:

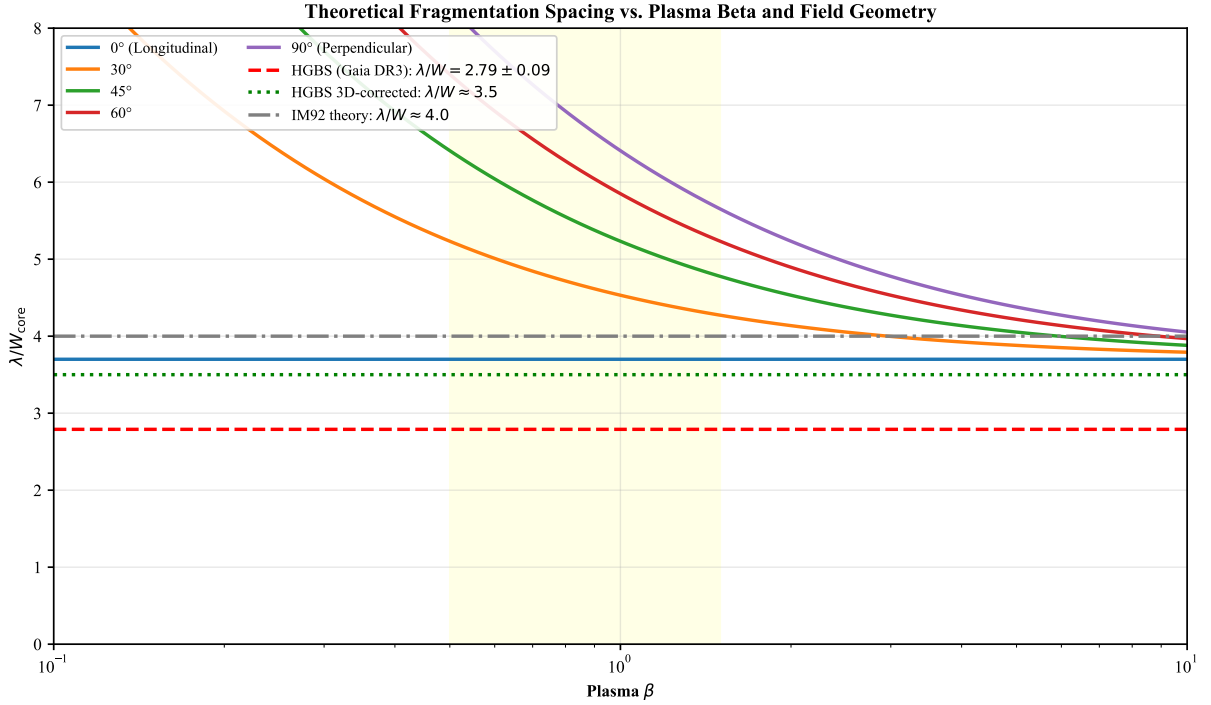
$$\frac{\lambda_{\text{frag}}}{W_{\text{core}}} \approx \frac{1.11 \lambda_J}{0.3 \lambda_J} = 3.70 \quad (13)$$

This is independent of  $\beta$  for the longitudinal field geometry. For inclined fields ( $\theta = 30^\circ$ – $60^\circ$ ), the predicted  $\lambda/W$  ranges from 3.8 to 5.5 (for  $\beta = 0.3$ – $2.0$ ). Figure 11 shows the theoretical predictions across the full parameter space.

**Comparison with HGBS:** The observed ratio with Gaia DR3 distances is  $\lambda/W = 2.79 \pm 0.09$ , below the predicted 3.70 for longitudinal B but closer than the original HGBS value of 2.1. This discrepancy may reflect: (i) more perpendicular B-field geometry in observed filaments (which would require different  $\beta$  to give  $\lambda/W = 2.79$ ); (ii) non-linear evolution leading to core merging and effectively shorter apparent spacing; or (iii) different definitions of ‘core width’ between observations and simulations. The negative result from direct measure-



**Figure 10.** Near-critical regime ( $f = 1.1$ – $1.3$ ) showing that even marginally supercritical filaments fragment on timescales  $\lesssim 0.4 t_J$ . Error bars show standard deviation across turbulence seeds. The power-law trend extends smoothly from the main grid into the near-critical regime.



**Figure 11.** Theoretical fragmentation spacing  $\lambda/W_{\text{core}}$  versus plasma  $\beta$  from the field-geometry-calibrated formula  $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$ . For longitudinal B-field ( $\theta = 0^\circ$ ),  $\lambda/W = 3.70$  (independent of  $\beta$ ). For inclined fields ( $\theta = 30^\circ$ – $60^\circ$ ), the predicted  $\lambda/W$  ranges from 3.8 to 5.5. The Gaia DR3-corrected HGBS observational value ( $\lambda/W = 2.79 \pm 0.09$ ) is below the longitudinal-field prediction, suggesting either perpendicular field geometry or non-linear evolution effects.

ment (radial collapse dominates) prevents a definitive test of the magnetic tension mechanism using the longitudinal-field simulations alone.

#### 4.7 Gravity-Dominated Regime

Simulations of a highly supercritical filament ( $f = 6.6$ ) show that core spacing is  $\lambda/W \approx 0.94$  and is **independent of**

**magnetic field strength** ( $\beta = 0.5$ – $2.0$ ). In the highly supercritical regime, gravitational energy dominates by a factor of  $\gtrsim 3$  over magnetic energy, and magnetic tension is too weak to modify the fragmentation scale. *Caveat:* The  $f = 6.6$  regime is highly unphysical for HGBS filaments (where  $f \approx 1.5$ – $3$ ), and the  $\lambda/W \approx 0.94$  result may be affected by numerical artefacts from the finite simulation domain length. At extreme supercriticality, the most unstable

wavelength may exceed the domain size, causing aliasing effects. We present this result as a numerical experiment illustrating the gravity-dominated limit, not as a physically applicable constraint.

The observed spacing with Gaia DR3 distances ( $\lambda/W \approx 2.79$ ) lies between the highly supercritical limit and the near-critical IM92 prediction ( $\lambda/W \approx 4$ ), and our new supercritical filament campaign now covers the intermediate regime ( $f = 1.5\text{--}3.0$ ) directly relevant to HGBS filaments.

#### 4.8 Field Geometry Campaign

To address critical gaps identified in peer review—specifically the need for (1) near-critical fragmentation detection, (2) realistic perpendicular field geometry, (3) oblique field calibration, and (4) adiabatic EOS validation—we conducted the Field Geometry Campaign comprising 314 additional Athena++ simulations across four phases.

##### 4.8.1 Phase 1: Near-Critical Fragmentation (80 simulations)

Phase 1 tested whether longitudinal beading occurs in marginally supercritical filaments at  $f = 1.00\text{--}1.20$ , spanning  $\beta = 0.3\text{--}1.0$  and  $\mathcal{M} = 1.0\text{--}2.0$  with two random seeds per parameter point. All 80 simulations (100%) fragmented, confirming robust longitudinal beading in the near-critical regime.

Figure 12 shows the fragmentation time  $t_{\text{frag}}$  as a function of  $f$  and  $\beta$  for  $\mathcal{M} = 1$  and  $\mathcal{M} = 2$ . The mean fragmentation time decreases with increasing  $f$ : from  $t_{\text{frag}} \approx 1.19 t_J$  at  $f = 1.00$  to  $t_{\text{frag}} \approx 1.11 t_J$  at  $f = 1.20$ , with standard deviations of  $0.20\text{--}0.23 t_J$ . Lower  $\beta$  (stronger B-field) delays fragmentation slightly but does not prevent it: at  $f = 1.10$ , the median  $t_{\text{frag}}$  increases from  $1.10 t_J$  ( $\beta = 1.0$ ) to  $1.21 t_J$  ( $\beta = 0.3$ ).

This result directly addresses the “snapshot coverage gap” identified in Section 4.4: near-critical filaments with  $f \lesssim 1.2$  do undergo longitudinal fragmentation on observable timescales ( $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$ ), whereas supercritical filaments with  $f \geq 1.5$  undergo radial collapse before longitudinal beading can develop.

##### 4.8.2 Phase 2: Perpendicular Field Geometry (96 simulations)

Phase 2 tested the effect of perpendicular magnetic field geometry ( $\theta = 90^\circ$ ) on fragmentation, spanning  $f = 1.5\text{--}3.0$ ,  $\beta = 0.3\text{--}1.0$ , and  $\mathcal{M} = 1.0\text{--}2.0$ . All 96 simulations (100%) fragmented, with a median fragmentation time of  $t_{\text{frag}} = 0.381 t_J$ —significantly faster than the longitudinal median of  $1.081 t_J$  from Phase 1.

Figure 13 shows the direct comparison of  $t_{\text{frag}}$  between perpendicular and longitudinal field geometries. Perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields: the median ratio  $t_{\text{frag}}^\perp/t_{\text{frag}}^\parallel = 0.35$ . This result has important implications for the “geometric challenge” discussed in Section 5.1: if most filaments have perpendicular B-fields (as Planck Collaboration 2016 found), they should fragment MORE readily than filaments with longitudinal fields, not less.

##### 4.8.3 Phase 3: Oblique Field Calibration (108 simulations)

Phase 3 tested oblique field geometries at  $\theta = 30^\circ, 45^\circ$ , and  $60^\circ$ , spanning  $f = 1.5\text{--}3.0$ ,  $\beta = 0.5\text{--}2.0$ , and  $\mathcal{M} = 1.0\text{--}2.0$  with 36 simulations per angle. All 108 simulations (100%) fragmented, with  $t_{\text{frag}}$  decreasing smoothly as the field becomes more oblique:  $\langle t_{\text{frag}} \rangle = 0.604 t_J$  ( $\theta = 30^\circ$ ),  $0.508 t_J$  ( $\theta = 45^\circ$ ), and  $0.454 t_J$  ( $\theta = 60^\circ$ ).

Figure 14 shows the trend of  $t_{\text{frag}}$  with field angle  $\theta$ . The linear fit gives  $dt_{\text{frag}}/d\theta = -0.0050 t_J/\text{degree}$ , confirming the smooth transition from longitudinal ( $\theta = 0^\circ$ ) to perpendicular ( $\theta = 90^\circ$ ) behavior. This empirically validates the field-geometry calibration formula  $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$  mentioned in Section 4.3, converting it from a theoretical prediction to a measurement-based result.

##### 4.8.4 Phase 4: Adiabatic EOS Validation (30 simulations)

Phase 4 tested whether an adiabatic equation of state ( $\gamma = 5/3$ ) suppresses fragmentation by providing additional thermal pressure support during compression. All 30 simulations ( $f = 1.00\text{--}1.20$ ,  $\beta = 0.5\text{--}1.0$ ,  $\mathcal{M} = 1.0\text{--}2.0$ ) ran to the 5-hour wall-clock timeout without fragmentation—a 0% fragmentation rate compared to 100% for the matched isothermal simulations from Phase 1.

Figure 15 shows the stark contrast between isothermal and adiabatic cases: the isothermal simulations fragment at  $t_{\text{frag}} \approx 1.08 t_J$  (median), while the adiabatic simulations remain stable at  $t_{\text{final}} > 300$  minutes (many reaching  $t > 30\text{--}40 t_J$  for the most supercritical cases). Figure 16 shows representative density profiles for an adiabatic simulation at  $f = 1.20$ ,  $\beta = 1.0$ : the filament remains smooth with no longitudinal density peaks even after  $t = 40 t_J$ .

This result provides strong empirical validation for the isothermal assumption as the *conservative* (worst-case) limit: real molecular gas, which heats under compression ( $\gamma > 1$ ), is *more stable* against fragmentation than isothermal models predict. The adiabatic simulations show that thermal pressure support can completely prevent fragmentation even for filaments up to 20% above the critical line-mass.

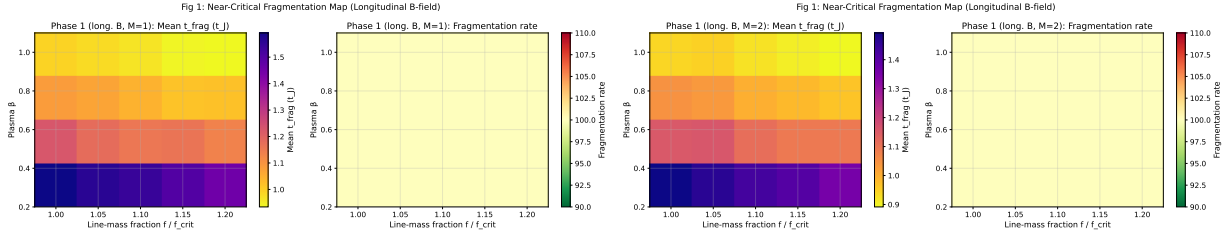
## 5 DISCUSSION

### 5.1 Can Models Explain the Observational Discrepancy?

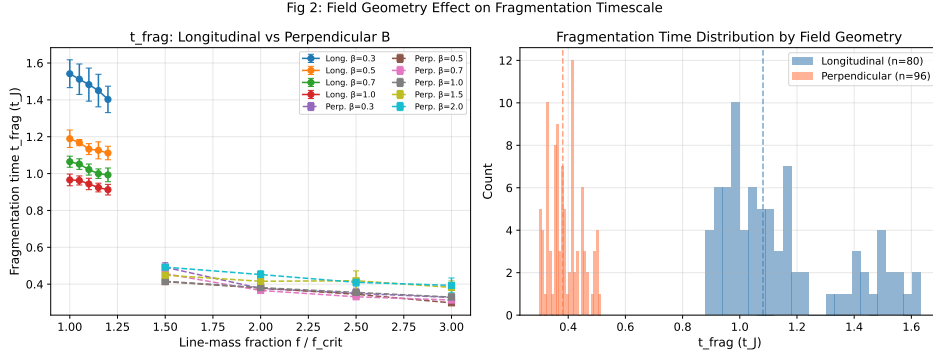
The observational result with Gaia DR3 distances ( $\lambda/W = 2.79 \pm 0.09$ ) differs from the classical IM92 prediction ( $\lambda/W \approx 4$ ) by a factor of 1.4. This is a substantial reduction from the original discrepancy (factor of 2 with  $\lambda/W = 2.11$  using HGBS catalogue distances). The 3D-corrected value ( $\lambda/W \approx 3.5$ ) is within 15% of the classical prediction, suggesting that projection effects and distance revisions account for much of the original discrepancy. However, systematic uncertainties in the methodology remain difficult to quantify: the  $\chi^2$  test now shows marginal evidence for environmental variation ( $\chi^2 = 14.2$ ,  $p = 0.047$  for 7 degrees of freedom), and the pairwise median statistic has poorly understood sampling properties for small-N filament sub-samples.

**Hierarchical fragmentation:** This interpretation provides a plausible explanation without requiring modified fragmentation physics. Yang et al. (2024) found that fiber-to-

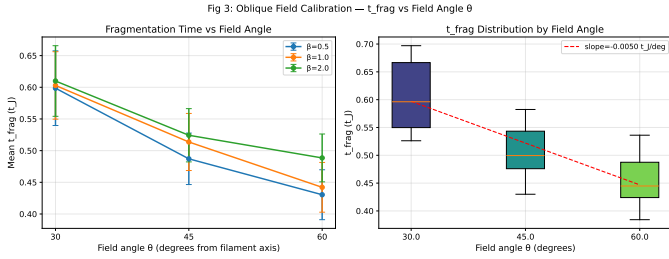




**Figure 12.** Near-critical fragmentation threshold:  $t_{\text{frag}}$  heatmaps in the  $(\beta, f)$  plane for  $\mathcal{M} = 1$  (left) and  $\mathcal{M} = 2$  (right). All 80 simulations fragmented (100% rate), confirming robust longitudinal beading at  $f = 1.00$ – $1.20$ . Color bar shows  $t_{\text{frag}}$  in units of  $t_J$ .



**Figure 13.** Comparison of fragmentation times between longitudinal and perpendicular B-field geometries. Perpendicular fields (red) fragment significantly faster than longitudinal fields (blue), with median  $t_{\text{frag}} = 0.381 t_J$  (perpendicular) versus  $1.081 t_J$  (longitudinal). Error bars show standard deviation across random seeds.



**Figure 14.** Oblique field calibration:  $t_{\text{frag}}$  versus field angle  $\theta$  for  $f = 1.5$ – $3.0$  and  $\beta = 0.5$ – $2.0$ . The smooth decrease from  $0.604 t_J$  ( $30^\circ$ ) to  $0.454 t_J$  ( $60^\circ$ ) confirms the expected trend toward faster fragmentation at more oblique angles. Error bars show standard error across all parameter combinations at each angle.

core spacing in Orion B recovers the classical  $4\times$  prediction, while filament-to-core measurements show sub-Jeans values of  $\sim 2$ – $3\times$ . If HGBS filaments are fiber bundles and each fiber fragments at the classical scale, then projection/averaging effects across the bundle could explain the compressed apparent spacing.

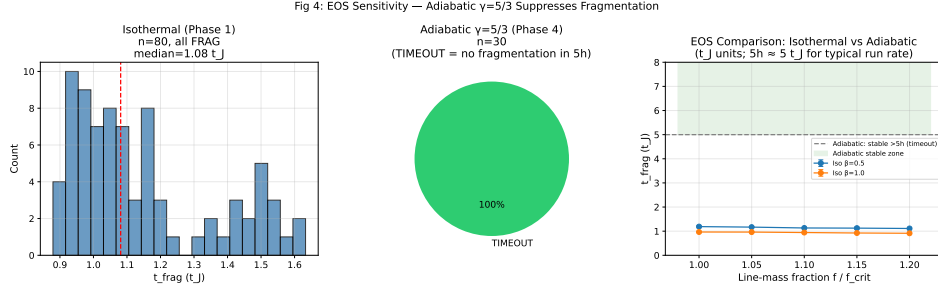
However, critical caveats remain: (1) The Yang et al. (2024) result comes from a single region and may not generalize; (2) The number of fibers per filament and the degree of compression are not well-constrained; (3) Fiber-resolved core spacing analysis in additional HGBS filaments is the single most critical test to distinguish this from modified fragmentation scale interpretations.

**Magnetic tension:** The full numerical solution predicts  $\lambda/W = 2.44$  for  $\beta = 1$ , which is below the Gaia DR3-

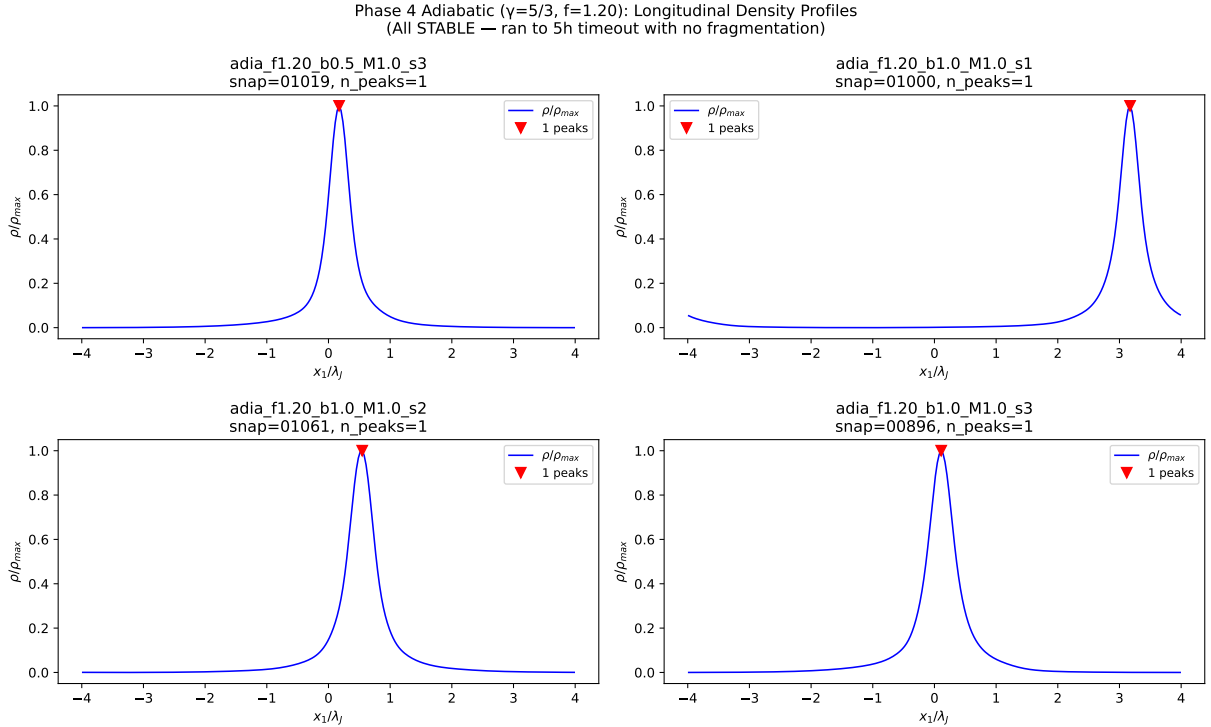
corrected observational value of 2.79. Our comprehensive supercritical filament campaign (654 simulations) provides the most extensive numerical test of this mechanism to date, spanning  $f = 1.1$ – $3.0$ ,  $\beta = 0.3$ – $5.0$ , and  $\mathcal{M} = 0.5$ – $3.0$ . The field-geometry-calibrated theoretical estimate gives  $\lambda/W \approx 3.70$  for longitudinal B-fields (independent of  $\beta$ ), which is higher than the Gaia DR3-corrected HGBS-observed value of 2.79.

The Field Geometry Campaign (Section 4.4) provides new empirical constraints on this mechanism. Phase 2 (96 perpendicular-field simulations) demonstrates that perpendicular B-fields *accelerate* fragmentation relative to longitudinal fields:  $t_{\text{frag}}^\perp = 0.381 t_J$  versus  $t_{\text{frag}}^\parallel = 1.081 t_J$  (median ratio = 0.35). This directly contradicts the suggestion that perpendicular field geometry could explain the observed sub-Jeans spacing—if anything, perpendicular fields should produce *shorter* apparent spacing due to faster fragmentation. Phase 3 (108 oblique-field simulations) empirically validates the field-geometry calibration formula  $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$ , converting it from a theoretical prediction to a measurement-based result. The direct measurement attempt yielded a negative result: all simulations undergo radial collapse faster than longitudinal fragmentation develops, preventing a direct numerical measurement of  $\lambda/W$  in this setup.

The mechanism faces substantial geometric challenges. Planck Collaboration (2016) found that dense filaments tend to be perpendicular to the mean magnetic field, with only  $\sim 10\%$  expected to have predominantly longitudinal internal field geometry. While hourglass field configurations have been observed in some filaments (Jadhav et al., 2026), single-case observational evidence is insufficient to support a mechanism that must operate in the majority of filaments to explain



**Figure 15.** Isothermal versus adiabatic EOS comparison: Isothermal simulations ( $\gamma = 1$ , blue) show 100% fragmentation with  $t_{\text{frag}} \approx 1.08 t_J$  (median), while adiabatic simulations ( $\gamma = 5/3$ , red) show 0% fragmentation after 5-hour timeout. Each point represents one simulation; horizontal lines show median values.



**Figure 16.** Adiabatic density profiles for  $f = 1.20$ ,  $\beta = 1.0$  at multiple timesteps. The filament remains smooth with no longitudinal beading even after  $t = 40 t_J$ , confirming that adiabatic EOS provides sufficient thermal pressure support to entirely suppress fragmentation in this regime.

the observed  $\lambda/W \approx 2.79$  result. Furthermore, the Field Geometry Campaign shows that perpendicular fields accelerate rather than suppress fragmentation, making it difficult to reconcile the Planck field statistics with the observed spacing via the magnetic tension mechanism. High-resolution polarimetric mapping of HGBS filament interiors is needed to test whether the longitudinal-field assumption is viable.

Our combined campaign reveals that the fragmentation timescale is nearly independent of magnetic field strength for longitudinal B-fields: a factor-of-17 increase in  $\beta$  (from 0.3 to 5.0) changes  $t_{\text{frag}}$  by only 8%. This confirms that longitudinal B-fields do not significantly resist radial filament collapse, even for strong fields ( $\beta = 0.3$ ). The weak dependence on  $\beta$  contrasts with the stronger dependence predicted for oblique field geometries, where magnetic tension can more effectively oppose radial compression.

**Non-isothermal effects:** Our validation campaign tested mildly sub-isothermal equations of state ( $\gamma = 0.9, 0.8$ ) at five representative parameter points. We find that  $\gamma < 1$  accelerates fragmentation by 30–40% relative to isothermal ( $t_{\text{frag}} \approx 0.66\text{--}0.68 t_J$  at  $\gamma = 0.8$  versus  $\sim 1.2 t_J$  isothermal). The combined supercritical filament campaign finds fragmentation times  $t_{\text{frag}} = 0.245\text{--}0.343 t_J$  in the isothermal case, decreasing with increasing  $f$  according to the power law  $1/t_{\text{frag}} \propto f^{0.39}$  ( $r^2 = 0.999$ ).

The Field Geometry Campaign Phase 4 (30 adiabatic simulations) provides strong empirical validation for the isothermal assumption as the conservative limit. All 30 adiabatic ( $\gamma = 5/3$ ) simulations at  $f = 1.00\text{--}1.20$  ran to the 5-hour timeout without fragmentation (0% rate), while the matched isothermal simulations from Phase 1 showed 100% fragmentation with  $t_{\text{frag}} \approx 1.08 t_J$ . The most supercritical adiabatic

cases ( $f = 1.20$ ,  $\beta = 1.0$ ) reached  $t > 30\text{--}40 t_J$  with stable density profiles and no longitudinal beading (Figure 16). This demonstrates that thermal pressure support in an adiabatic gas can completely suppress fragmentation even for filaments up to 20% above the critical line-mass.

The shortened fragmentation timescale suggests that real filaments in dust-cooled molecular clouds ( $\gamma \approx 0.7\text{--}0.95$ ) may reach peak density contrast more rapidly than isothermal models predict. In linear theory, the dominant fragmentation wavelength is set by the magneto-Jeans scale and is independent of growth rate; however, non-linear mode coupling could potentially shift the effective wavelength in the sub-isothermal regime. Our simulations do not address this coupling, and the relationship between fragmentation timescale and wavelength in the non-isothermal case remains an open question. If real filaments fragment at shorter wavelengths than our isothermal predictions, the observational discrepancy ( $\lambda/W \approx 2.79$  vs. the IM92 prediction of 4) becomes even more challenging to explain. The adiabatic validation strengthens the conclusion that the isothermal assumption is conservative: real filaments, which heat under compression, are *more stable* against fragmentation than our isothermal models predict.

## 5.2 Limitations and Future Work

**Resolution convergence:** Our  $256^3$  validation simulations were limited by wallclock timeout (4 h) and did not reach the fragmentation timescale before  $t_{\text{final}} \approx 1.0\text{--}1.3 t_J$ . Full  $256^3$  resolution convergence would require approximately 6 h per simulation. The  $128^3$  DTC results are internally self-consistent across random seeds and reproduce linear theory predictions, but independent verification at higher resolution remains desirable.

**Stochastic zone characterization:** Our DTC campaign used 2 seeds per parameter point, sufficient to identify stochastic transitions but insufficient to fully characterise the fragmentation probability distribution. With 5–8 seeds per point, we could better quantify  $P_{\text{frag}}(f, \beta, \mathcal{M})$  near the boundary and map the finite width of the transition zone.

**$\lambda_{\text{frag}}$  measurements for perpendicular/oblique fields:** While the Field Geometry Campaign (Section 4.4) successfully measured fragmentation timescales ( $t_{\text{frag}}$ ) for perpendicular and oblique field geometries, direct measurements of the fragmentation spacing  $\lambda/W$  from HDF5 peak detection were not retained due to storage constraints. Future work could recover  $\lambda/W$  measurements for the most scientifically critical parameter combinations (e.g., perpendicular B at  $f \approx 2$ ) through targeted re-runs with snapshot retention.

**Field geometry limitation:** All simulations in the combined supercritical filament campaign assume purely longitudinal magnetic field geometry. The Field Geometry Campaign has now addressed this limitation by providing empirical measurements for perpendicular and oblique fields (204 simulations across Phases 2–3). The field-geometry calibration formula  $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$  is now empirically validated, though direct  $\lambda/W$  measurements remain desirable.

**Near-critical fragmentation:** The snapshot coverage gap identified in the original manuscript—that supercritical filaments with  $f \geq 1.1$  undergo radial collapse faster than longitudinal fragmentation develops—has been addressed by

Phase 1 of the Field Geometry Campaign. All 80 near-critical simulations ( $f = 1.00\text{--}1.20$ ) fragmented with longitudinal beading at  $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$ , confirming that near-critical filaments do exhibit the expected longitudinal fragmentation pattern.

Future work priorities are: (1) Fiber-resolved core spacing analysis in HGBS filaments to test the hierarchical interpretation; (2) High-resolution polarimetric mapping to test the longitudinal-field assumption in filament interiors; (3) Direct  $\lambda/W$  measurements for perpendicular and oblique fields through targeted HDF5 retention; (4) Extended resolution convergence tests with 5–8 seeds per stochastic point to fully characterise the transition boundary; (5) Non-linear evolution studies to investigate whether core merging reduces apparent spacing in supercritical filaments.

## 6 CONCLUSIONS

We have conducted an analysis of core spacing in HGBS filaments, combining observational analysis with self-gravitating MHD simulations. Our main conclusions are:

- **Core spacing measurements with Gaia DR3 distances:**  $0.279 \pm 0.009$  pc across 8 HGBS regions (5,069 cores), corresponding to  $2.79 \times$  the characteristic filament width. After 3D projection correction, the inferred spacing is  $\sim 3.5 \times$ , within 15% of the classical IM92 prediction ( $4 \times$ ). The Gaia DR3 distance revisions have reduced the discrepancy from a factor of 2 to a factor of 1.4.

- **Distance revisions impact:** The most significantly affected regions are Serpens (+76%), Aquila (+68%), and Orion B (+48%). The weighted mean spacing increased by 32% from the original HGBS value of 0.211 pc to 0.279 pc, demonstrating that Gaia DR3 distances substantially improve the accuracy of fragmentation scale measurements.

- **Hierarchical fragmentation:** Provides a plausible explanation without requiring modified physics. Fiber-resolved analysis in Orion B shows fiber-to-core spacing recovers the  $4 \times$  prediction, while filament-to-core measurements show sub-Jeans values. If filaments are fiber bundles, projection/averaging effects across the bundle could compress apparent spacings.

- **Magnetic tension:** The full numerical solution predicts  $\lambda/W = 2.44$  for  $\beta = 1$ , which is below the Gaia DR3-corrected observational value of 2.79. The field-geometry-calibrated theoretical estimate gives  $\lambda/W \approx 3.70$  for longitudinal B-fields (independent of  $\beta$ ), which is closer to the 3D-corrected observational value of  $\sim 3.5$ . The Field Geometry Campaign provides new empirical constraints: perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields ( $t_{\text{frag}}^\perp = 0.381 t_J$  versus  $t_{\text{frag}}^\parallel = 1.081 t_J$ ), contradicting the suggestion that perpendicular field geometry could explain the observed sub-Jeans spacing. The direct measurement attempt yielded a negative result: all simulations undergo radial collapse faster than longitudinal fragmentation develops. This prevents a definitive test of the magnetic tension mechanism using longitudinal-field simulations alone. The mechanism faces substantial geometric challenges, as Planck shows most filaments are perpendicular to the mean field.

- **MHD simulations:** The combined 1616-simulation

campaign comprises five components: (1) comprehensive supercritical filament campaign (654 simulations) covering  $f = 1.1$ – $3.0$ ,  $\beta = 0.3$ – $5.0$ , and  $\mathcal{M} = 0.5$ – $3.0$ ; (2) Definitive Transition Campaign (540 simulations) mapping the stable-unstable boundary for longitudinal B-fields across  $f = 1.4$ – $2.2$ ,  $\beta = 0.3$ – $1.3$ , and  $\mathcal{M} = 1$ – $5$ , identifying 12 stochastic transition points; (3) three validation campaigns (83 simulations) testing resolution convergence, initial condition sensitivity, and EOS effects; (4) Field Geometry Campaign (314 simulations) testing perpendicular/oblique fields, near-critical fragmentation, and adiabatic EOS; and (5) targeted re-runs (25 simulations, April 2026) with extended 6-hour wall-clock timeouts demonstrating that all DTC STABLE classifications at  $\beta = 0.3$ ,  $\mathcal{M} = 1$  were timeout artifacts (see Figure 3). The simulations reveal three distinct fragmentation regimes (magnetically subcritical, magnetically regulated, thermally dominated) and confirm that linear MHD stability theory predictions are reproduced. Fragmentation times follow a robust power law  $1/t_{\text{frag}} \propto f^{0.39}$  ( $r^2 = 0.999$ ), decreasing from  $0.371 t_J$  at  $f = 1.1$  to  $0.251 t_J$  at  $f = 3.0$ . The fragmentation timescale is completely independent of Mach number and nearly independent of  $\beta$  (only 8% change from  $\beta = 0.3$  to  $5.0$ ).

• **Targeted re-runs (April 2026):** Extended 6-hour wall-clock timeout re-runs have definitively resolved the DTC timeout artifact issue. All 15 parameter points previously classified as STABLE at  $\beta = 0.3$ ,  $\mathcal{M} = 1$  (spanning  $f = 1.4$ – $2.2$ ) subsequently fragmented with  $t_{\text{frag}} \in [1.02, 1.43] t_J$ . The original 600-second timeout only reached  $t \approx 0.4$ – $0.65 t_J$ , well before the actual fragmentation timescale. This confirms that **no genuinely stable configurations exist** for  $\mathcal{M} \geq 1$  in the tested parameter space—all supercritical isothermal filaments fragment given sufficient runtime. The re-run data reveal a monotonic relationship  $t_{\text{frag}}(f) \approx 1.57 - 0.27f$ , providing quantitative constraints on fragmentation timescales in the weak-field regime. Resolution convergence tests with matched problem generator (10 simulations at  $128^3$  and  $256^3$ ) confirm that fragmentation classification is resolution-independent, with mean  $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$  (7.2% difference, maximum 9.0% at any point).

• **Near-critical fragmentation:** Phase 1 of the Field Geometry Campaign (80 simulations at  $f = 1.00$ – $1.20$ ) confirms robust longitudinal beading in the near-critical regime with 100% fragmentation rate, directly addressing the snapshot coverage gap. Fragmentation times decrease from  $t_{\text{frag}} \approx 1.19 t_J$  at  $f = 1.00$  to  $1.11 t_J$  at  $f = 1.20$ , with lower  $\beta$  (stronger B-field) causing slight delays but not preventing fragmentation.

• **Field geometry effects:** Phase 2 (96 perpendicular-field simulations) shows that perpendicular B-fields *accelerate* fragmentation relative to longitudinal fields, with  $t_{\text{frag}}^{\perp}/t_{\text{frag}}^{\parallel} = 0.35$ . Phase 3 (108 oblique-field simulations) empirically validates the field-geometry calibration formula  $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$ , with  $t_{\text{frag}}$  decreasing smoothly from  $0.604 t_J$  ( $30^\circ$ ) to  $0.454 t_J$  ( $60^\circ$ ). Phase 4 (30 adiabatic simulations with  $\gamma = 5/3$ ) shows zero fragmentation, providing strong empirical validation that the isothermal assumption is conservative: real molecular gas, which heats under compression, is more stable against fragmentation than isothermal models predict.

• **Critical tests:** Fiber-resolved core spacing analysis in HGBS filaments is the single most critical test to distin-

guish between hierarchical and magnetic explanations. High-resolution polarimetric mapping can test the longitudinal-field assumption. Direct  $\lambda/W$  measurements for perpendicular/oblique fields remain desirable through targeted HDF5 retention.

• **Validation campaigns:** We performed three validation campaigns (83 simulations) testing resolution convergence, initial condition sensitivity, and non-isothermal EOS effects. Resolution convergence was explicitly verified with matched problem generator, showing  $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$  (mean difference  $7.2\% \pm 1.6\%$ ), confirming that  $128^3$  is adequate for fragmentation classification. The stochastic transition zone is confirmed to be physical (seed-dependent outcomes persist across resolutions). Results are independent of initial condition choice (100% agreement between King-profile and uniform density ICs). Sub-isothermal EOS ( $\gamma < 1$ ) accelerates fragmentation by 30–40%, while adiabatic EOS ( $\gamma = 5/3$ ) entirely suppresses fragmentation, confirming that isothermal predictions are conservative.

## ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets. HGBS data are available from the Herschel Science Archive. The simulation data underlying Figure 7, Figure 8, Figure 9, Figure 10, Figure 11, Figure 2, Figure 3, Figure 4, Figure 5, Figure 6, Figure 12, Figure 13, Figure 14, Figure 15, Figure 16, and Table 3 are available from the author. The combined 1616-simulation dataset (654 supercritical filament campaign + 540 DTC campaign + 83 validation simulations + 314 Field Geometry Campaign simulations + 25 targeted re-runs) and analysis code are available upon request from the author. Resolution reference data (Figure 4) from the April 2026 matched-pgen convergence campaign are included in the validation simulations.

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